



Molecular detection of equine respiratory viruses in horses from Algeria: first evidence of equine rhinitis B virus circulation

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INTRODUCTION

Globally, equids are particularly susceptible to respiratory diseases, which represent an important cause of morbidity and reduced performance in horses (Dunowska et al., 2002).

Equine rhinoviruses are among the less well-characterized viral pathogens associated with respiratory disease and reduced performance in horses (Back et al., 2019).

Two distinct types have been identified in equids, namely equine rhinitis A virus (ERAV) and equine rhinitis B virus (ERBV) (Horsington et al., 2013). Both viruses have been detected worldwide in horses, including clinically healthy animals and those presenting with respiratory disease (Diaz-Mendez et al., 2010).

Despite their worldwide distribution, information on the circulation of equine rhinoviruses and other major respiratory viruses in Algerian horses remains limited.

Therefore, this study aimed to investigate the molecular occurrence of major equine respiratory viruses, with particular emphasis on the detection of ERAV and ERBV, in horses from Algeria.

MATERIALS AND METHODS

Sample collection

A total of 97 clinical samples from 72 horses with respiratory signs were collected in Algeria (Figure 1) between May 2022 and June 2025: 73 nasopharyngeal swabs (NS), 21 fecal and 3 urine samples. Samples were stored at -80 °C and analyzed at the WOAHP Reference Laboratory, Irish Equine Centre

Molecular detection

ERAV and ERBV were investigated in all samples, while EIV, EHV-1 and EHV-4 were additionally tested in 50 NS. Viral nucleic acids were extracted using the MagMAX Core kit/KingFisher Flex system. Detection was performed by real-time PCR/RT-PCR using validated assays. ERAV/ERBV assays targeted the 5' UTR (Quinlivan et al., 2010), and EIV the matrix gene (Heine et al., 2007). Primers and probes are presented in Table 1.

Virus	Forward primer (5' to 3')	Reverse primer (5' to 3')	Probe sequence (5' to 3')
EHV-1	GCGGGCTC TGACAACA CAA	TTGTGGTT TCATGGGA GTGTGTA	6-FAM-T AACGCA AACGGT ACAGA A- MGB
EHV-4	TGTTGCCG TGCTATYG TG	GCTGGAGT ACGGTGTA TAGAA	Cy5- TTATTGGC GAAGAGA CTGTGGTA- BHQ3
EIV	AGATGAGY CTTCTAAC CGAGGTCG	TGCAAANA CATCYTCA AGTCTCTG	6-FAM-T CAGGCC CCCTCA AAGCCG A- TAMRA
ERAV	CCAGGTAA CCGGACAG CG	GCGAGCGC TACCACAG G	6-FAM-C ATTGCT CTGGA TTGGT T- MGB
ERBV	TGATGCTT GGCTCTCA GAAA	GCAAACG ACCAAACA TCAA	6-FAM-CTT CCA ACT AAA CCC- MGB

Table 1. Primers and probes used for the molecular detection of equine respiratory viruses

Virus	Sampling period	Samples tested	Results
ERAV	May 2022–October 2023	23 NS	0 positive
ERBV	May 2022–October 2023	23 NS	0 positive
ERAV	November 2023–June 2025	50 NS + 21 fecal + 3 urine	0 positive
ERBV	November 2023–June 2025	50 NS + 21 fecal + 3 urine	2 positive NS
EIV	November 2023–June 2025	50 NS	0 positive
EHV-1	November 2023–June 2025	50 NS	0 positive
EHV-4	November 2023–June 2025	50 NS	3 positive NS

Table 2. Molecular detection of equine respiratory viruses in clinical Samples collected from horses in Algeria

CONCLUSION

The detection of ERBV and EHV-4, including two cases of co-infection in horses from Sétif, provides evidence of the circulation of these respiratory viruses in Algeria. Further epidemiological and molecular investigations are needed to better characterize their distribution and potential role in equine respiratory disease.

RESULTS

All samples collected between May 2022 and October 2023, including 23 nasopharyngeal swabs (NS), tested negative for ERAV and ERBV. During the second sampling period, from November 2023 to June 2025, all 21 fecal samples and 3 urine samples were negative for ERAV and ERBV.

Among the 50 NS collected during this period and screened for ERAV, ERBV, EIV, EHV-1, and EHV-4, ERAV and EHV-4 were detected in three horses.

ERBV was detected in two NS (2.8%), while EHV-4 was detected in three NS (4.2%). Two horses from Sétif were simultaneously positive for ERBV and EHV-4, indicating co-infection. The third EHV-4-positive horse, without ERBV co-detection, originated from Batna. All samples tested negative for ERAV, EIV, and EHV-1.

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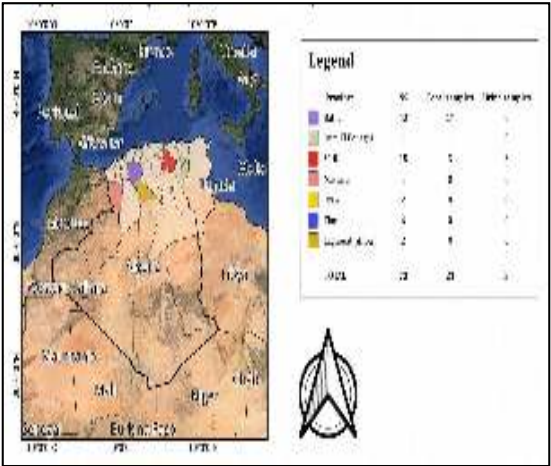


Figure 1 : Geographic distribution and origin of samples collected from horses in Algeria